LESSON 2.7

Divide Decimals Using an Algorithm

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide with decimals by making the divisor a whole number first.

WHY THIS MATTERS A school bought new math workbooks as shown in the order form. What is the cost of each workbook?

REMEMBER

Divide Decimals Using an Algorithm

Decimal Division Algorithm Dividing splits a total into equal groups. When you divide by a decimal, first move the point to make the divisor a whole number — then run the same long-division cycle: divide, multiply, subtract, bring down.

1. Make the divisor a whole number: slide its decimal point to the right.
2. Slide the dividend's decimal point to the right the same number of places.
3. Write the answer's decimal point in the column directly above the dividend's point, then divide as whole numbers using Divide → Multiply → Subtract → Bring Down.

Words: **Dividend *(Dividendo)*** **Divisor *(Divisor)*** **Quotient *(Cociente)*** **Decimal Division Algorithm *(Algoritmo de división decimal)*** **Equivalent division *(División equivalente)***

PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

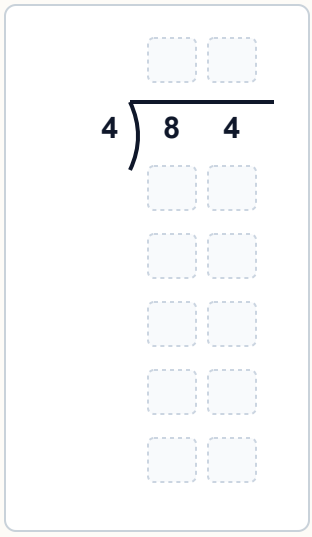
What is 84 ÷ 4?

(A) 12

(B) 20

(C) 21

(D) 336



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

2 **Circle the letter of the best answer. Show how you know.**

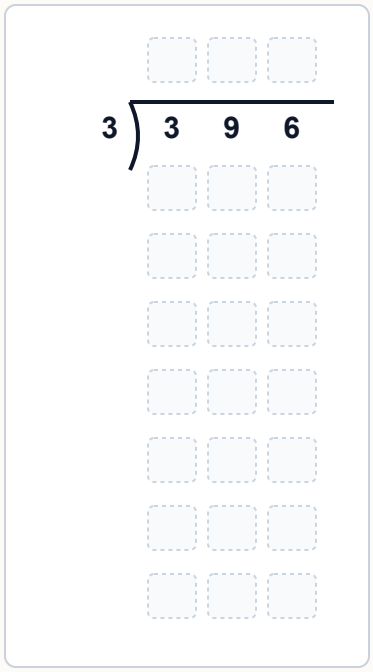
What is 396 ÷ 3?

(A) 13.2

(B) 122

(C) 132

(D) 133

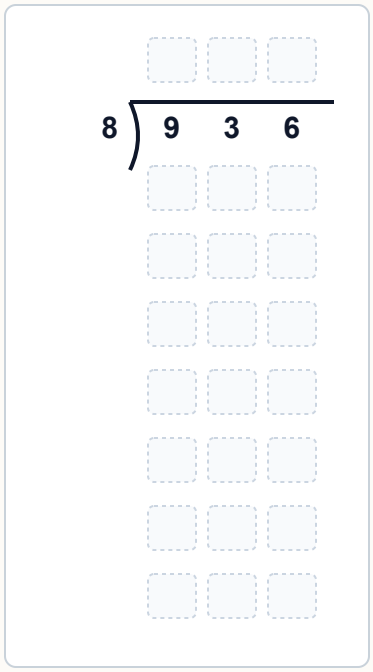


1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

3 **Circle the letter of the best answer. Show how you know.**

What is 936 ÷ 8?

(A) 107
(B) 116
(C) 117
(D) 127



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A baker has 50 rolls. Each box holds 6 rolls. How many boxes are filled, and how many rolls are left over?

(A) 6 boxes, 2 left over
(B) 8 boxes, 2 left over
(C) 8 boxes, 4 left over
(D) 9 boxes, 4 left over

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A ribbon is 16.8 feet long. You need to cut it into pieces that are each 2.4 feet long for a craft project. 16.8 ÷ 2.4 = ?

(A) 0.7
(B) 4.2
(C) 7
(D) 70

SHOW YOUR WORK

6 Circle the letter of the best answer. Show how you know.

A ribbon is 16.8 feet long. You need to cut it into pieces that are each 2.4 feet long for a craft project. Why can you rewrite 16.8 ÷ 2.4 as 168 ÷ 24?

(A) Because decimals are not allowed in division
(B) Because 168 and 16.8 are equal
(C) Because you always drop the decimal point
(D) Multiplying both numbers by 10 keeps the quotient the same

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does moving the point the same number of places in both numbers give you the same quotient — and why is it worth doing before you start the divide, multiply, subtract, bring down cycle?

SENTENCE STARTER Multiplying both numbers by ___ doesn't change the quotient because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 4.8 ÷ 0.6?

(A) 0.8
(B) 4.2
(C) 8
(D) 80

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is 9.45 ÷ 0.9?

(A) 1.05
(B) 9.5
(C) 10.5
(D) 105

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 15 ÷ 0.25?

(A) 0.6
(B) 1.5
(C) 6
(D) 60

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A ribbon is 16.8 feet long. You need to cut it into pieces that are each 2.4 feet long for a craft project. If each piece were 2.8 feet instead, how many whole pieces could you cut?

(A) 5
(B) 6
(C) 7
(D) 8

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A ribbon is 7.2 meters long and is cut into 0.8-meter pieces. How many pieces are there?

(A) 7
(B) 8
(C) 9
(D) 10

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Table 1's lunch bill is $15.00. Every sandwich costs $3.75. How many sandwiches did Table 1 order?

(A) 0.25 sandwiches

(B) 4 sandwiches

(C) 11.25 sandwiches

(D) 40 sandwiches

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does moving the point the same number of places in both numbers give you the same quotient — and why is it worth doing before you start the divide, multiply, subtract, bring down cycle?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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